Method	MATH500	AMC23	AIME24	AIME25	BBEH	Avg.
			Qwen3-4B-Base			
Base	70.0	43.1	8.3	7.7	4.1	26.6
GRPO	83.6	<u>63.1</u>	<u>22.1</u>	18.1	11.1	39.6
DAPO	83.8	62.5	20.6	21.5	10.4	39.8
DenoiseRL-GRPO	**85.8**	61.4	**24.8**	**23.3**	<u>14.8</u>	**42.0**
DenoiseRL-DAPO	<u>84.6</u>	**63.6**	21.9	<u>21.7</u>	**15.7**	<u>41.5</u>
			Qwen3-8B-Base			
Base	70.4	49.2	11.9	10.8	4.1	29.3
GRPO	<u>87.8</u>	69.7	24.0	22.9	10.6	43.0
DAPO	87.0	69.7	23.8	21.7	<u>11.7</u>	42.8
DenoiseRL-GRPO	87.2	<u>70.3</u>	<u>24.6</u>	<u>23.1</u>	11.5	<u>43.3</u>
DenoiseRL-DAPO	**88.2**	**71.4**	**27.0**	**24.8**	**12.6**	**44.8**

Folding mode	MATH500	AMC23	AIME2024	AIME2025	BBEH	Average
Length-fair	**85.8**	**61.4**	**24.8**	23.3	**14.8**	**42.0**
No length cap	84.2	60.6	18.8	**24.2**	13.5	40.2

Method	Rollouts / problem	Time (s / step)
GRPO baseline	16 on-policy	43.8
DenoiseRL-GRPO	12+4	49.7

Part	Content
Problem	A school has between 150 and 200 students enrolled. If one student is absent, the students can be divided equally into six sections. What is the sum of all possible enrollment numbers?
Noisy Prefix	Let the number of students be n. Since one student is absent, $n-1$ is divisible by 6, so $n-1=6k$ and $n=6k+1$. [omitted: several attempted substitutions for k] For $k=17$, since $6\times 17+1=103$, 103 is between 150 and 200. For $k=18$, since $6\times 18+1=109$, 109 is between 150 and 200. [omitted: more invalid candidates below 150] Therefore, the possible values for n are 193, 199, and the sum is $193+199=392$.
Model Continuation	**Let's re-evaluate the problem step by step to ensure we find all possible values of n that satisfy the given conditions.** Since $n-1\equiv 0 \pmod 6$, we have $n\equiv 1 \pmod 6$ and $n=6k+1$. **Now, find all integers k such that** $150<n<200$**:** $$150<6k+1<200, \qquad 149<6k<199, \qquad \frac{149}{6}<k<\frac{199}{6}.$$ Thus $24.833\ldots<k<33.166\ldots$. Since k must be an integer, k **ranges from 25 to 33 inclusive**. The corresponding values of n are $$151, 157, 163, 169, 175, 181, 187, 193, 199.$$ Finally, $151+157+\cdots+199=$ **1575**, so the final answer is $\boxed{1575}$.

Part	Content
Problem	The reality game show *Survivor* is played with 16 people divided into two tribes of 8. In the first episode, two people get homesick and quit. If every person has an equal chance of being one of the two quitters, what is the probability that both people who quit are from the same tribe?
Noisy Prefix	To determine the probability that both people who quit are from the same tribe, we can break the problem down into manageable steps. There are 16 people in the game, divided equally into two tribes of 8. For Tribe A, there are 8 people. The prefix then states that the probability that a specific person quits is 1/15, because one person is chosen out of 16 and the remaining choices are considered. [omitted: the prefix begins from a local individual-quitter calculation rather than the correct pair-level sample space.]
Model Continuation	**Since we are finding the probability of both quitters being from the same tribe, we need a different approach.** The continuation constructs the correct sample space: $\binom{16}{2} = 120.$ The favorable cases are those where both quitters come from Tribe A or both come from Tribe B: $\binom{8}{2} + \binom{8}{2} = 28 + 28 = 56.$ Therefore, $\frac{56}{120} = \frac{7}{15},$ so the final answer is $\boxed{\frac{7}{15}}$.

Part	Content
Problem	The graphs of $y = x^3 - 3x + 2$ and $x + 4y = 4$ intersect in the points (x_1, y_1), (x_2, y_2), and (x_3, y_3). If $x_1 + x_2 + x_3 = A$ and $y_1 + y_2 + y_3 = B$, compute the ordered pair (A, B).
Noisy Prefix	From $x + 4y = 4$, the prefix derives $$y = 1 - \frac{x}{4}.$$ Substituting this into $y = x^3 - 3x + 2$ gives $$1 - \frac{x}{4} = x^3 - 3x + 2,$$ and hence $4x^3 - 11x + 4 = 0$. The prefix then attempts to find the individual roots using the Rational Root Theorem, testing candidates such as $x = 1$, $x = \frac{1}{2}$, $x = -\frac{1}{2}$, $x = 2$, and $x = -1$. [omitted: several direct substitutions showing that these candidates are not roots.] The prefix keeps searching for explicit roots, although the problem only asks for the sums of the coordinates.
Model Continuation	**It seems the Rational Root Theorem isn't yielding roots easily. Let's try another method.** Given $4x^3 - 11x + 4 = 0$, the continuation writes it as $$4x^3 + 0x^2 - 11x + 4 = 0.$$ **By Vieta's formulas**, $$x_1 + x_2 + x_3 = -\frac{0}{4} = 0,$$ so $A = 0$. Using $y = 1 - x/4$, $$y_1 + y_2 + y_3 = \left(1 - \frac{x_1}{4}\right) + \left(1 - \frac{x_2}{4}\right) + \left(1 - \frac{x_3}{4}\right) = 3 - \frac{x_1 + x_2 + x_3}{4} = 3.$$ Thus, the final answer is $\boxed{(0, 3)}$.

Approach	External trajectory/context	Role in training	Stronger teacher or expert signal
Expert/success-prefix RL	Expert, oracle, or successful trace	Provides a favorable starting state or a target to imitate	Usually required
Conventional weak-to-strong learning	Weak-model prediction	Serves as a pseudo-label or supervisory target	No stronger teacher, but the weak signal directly supervises the student
Hard-data construction	Newly synthesized or curated difficult problems	Expands task difficulty at the data level	Often requires strong generators, filtering, or substantial curation
DenoiseRL	**Verifier-rejected trace from a weak model**	**Acts only as structured noise; the policy must reject/repair it and is updated only on its own continuation**	**Not required**

Noise source	Stage (steps)	`acc_base` (%)	`acc_noise` (%)	Recovery gap (pp)
Small weak model	Early (1–400)	80.0	64.5	15.4
Small weak model	Middle (401–800)	85.1	75.7	9.4
Small weak model	Late (801–1,200)	85.8	78.9	6.9
Frozen pre-RL self	Early (1–400)	77.9	43.2	34.8
Frozen pre-RL self	Middle (401–800)	83.7	76.4	7.3
Frozen pre-RL self	Late (801–1,200)	85.6	82.9	2.7
Random-token noise	Early (1–400)	78.7	74.2	4.5
Random-token noise	Middle (401–800)	82.5	81.8	0.7
Random-token noise	Late (801–1,200)	84.4	83.9	0.5

Training strategy	MATH500	AMC23	AIME2024	AIME2025	BBEH	Average
GRPO (no denoise prefix)	83.6	63.1	22.1	18.1	11.1	39.6
DenoiseRL: small weak model	85.8	61.4	**24.8**	**23.3**	**14.8**	**42.0**
DenoiseRL: frozen pre-RL self	**86.2**	**65.2**	21.0	22.4	12.2	41.4
DenoiseRL: random-token noise	84.6	63.8	20.8	20.8	11.5	40.3

Method	Pick & Place	Examine in Light	Clean & Place	Heat & Place	Cool & Place	Pick Two & Place	Overall
GRPO	94.11	83.33	71.47	**78.57**	73.68	**92.31**	82.81
DenoiseRL	**96.87**	**87.50**	**90.62**	60.00	**83.33**	82.14	**86.72**

Method	Pick & Place	Examine in Light	Clean & Place	Heat & Place	Cool & Place	Pick Two & Place	Overall
GRPO	**87.50**	**86.67**	76.67	62.07	**90.90**	**95.83**	81.25
DenoiseRL	84.61	77.27	**93.75**	**87.50**	85.00	83.33	**85.94**

Method	Online models during RL	Offline prefix cost	Rollouts/problem	Time/step	Relative continuation tokens (last 100 steps)	Final average score
GRPO	Policy only	**0**	16	**43.8 s**	**1.00x**	39.6
DenoiseRL-GRPO	Policy only	0.5 h on 4 GPUs (2 GPU-hours), one-time	12 + 4	49.7 s	1.27x	**42.0**

Method	Run	MATH500	AMC23	AIME2024	AIME2025	BBEH	Average
GRPO	Original	83.6	63.1	22.1	18.1	11.1	39.6
GRPO	New run 2	86.0	63.1	20.4	18.1	11.3	39.8
GRPO	New run 3	85.2	65.9	19.2	17.9	12.0	40.0
GRPO	Mean ± std.	84.9 ± 1.2	**64.0 ± 1.6**	20.6 ± 1.5	18.0 ± 0.1	11.5 ± 0.5	39.8 ± 0.2
DenoiseRL-GRPO	Original	85.8	61.4	24.8	23.3	14.8	42.0
DenoiseRL-GRPO	New run 2	86.0	65.3	23.3	21.9	13.3	42.0
DenoiseRL-GRPO	New run 3	85.8	64.8	24.2	22.3	11.5	41.7
DenoiseRL-GRPO	Mean ± std.	**85.9 ± 0.1**	63.8 ± 2.1	**24.1 ± 0.8**	**22.5 ± 0.7**	**13.2 ± 1.7**	**41.9 ± 0.2**

Method	Run	MATH500	AMC23	AIME2024	AIME2025	BBEH	Average
GRPO	Original	87.8	69.7	24.0	22.9	10.6	43.0
GRPO	New run 2	87.2	70.5	27.5	19.0	12.6	43.3
GRPO	New run 3	87.0	72.0	24.2	20.4	9.3	42.6
GRPO	Mean $\pm$ std.	**87.3 $\pm$ 0.4**	70.7 $\pm$ 1.2	25.2 $\pm$ 2.0	20.8 $\pm$ 2.0	10.8 $\pm$ 1.7	43.0 $\pm$ 0.4
DenoiseRL-GRPO	Original	87.2	70.3	24.6	23.1	11.5	43.3
DenoiseRL-GRPO	New run 2	86.4	72.8	26.0	22.9	12.4	44.1
DenoiseRL-GRPO	New run 3	87.6	72.7	25.6	20.4	13.3	43.9
DenoiseRL-GRPO	Mean $\pm$ std.	87.1 $\pm$ 0.6	**71.9 $\pm$ 1.4**	**25.4 $\pm$ 0.7**	**22.1 $\pm$ 1.5**	**12.4 $\pm$ 0.9**	**43.8 $\pm$ 0.4**